

SdCvP-X™: Electronic (Design) Verification Automation Tool

VERIPOINT TECHNOLOGIES PVT LTD

AI Tools & Solutions Powering Semiconductors and Beyond

Company Profile

Veripoint Technologies is a deep-tech product company addressing critical industry challenges, from automating semiconductor verification to delivering scalable, intelligent automation for industrial sectors through its AI-powered tools and engineering solutions.

Industry Challenges

- **Semiconductors:** Manual verification slows development and increases the risk of costly silicon failures.
- **Industrial Sectors:** Scalable, intelligent automation solutions are still limited, hindering operations, efficiency and innovation.

Solutions: Our Flagship Products & Premium DV Services

- **SdCvP-X™** : Proprietary AI-driven Electronic (Design) Verification automation tool.
- **AI-driven solutions** for manufacturing, automotive, telecommunications, healthcare and more.
- **Premium DV services** delivered by a team of experts with deep specialization in verifying complex designs.

Founded: Feb 2022 | **HQ:** Noida, UP, India | **Website:** www.veripointtech.com | **DPIIT:** DIPP158213

SdCvP-X™: Electronic (Design) Verification Automation Tool

SdCvP-X™ is our proprietary AI tool that automates chip design verification, accelerating development and ensuring thorough verification.

Target Designs

- IP (standard & custom), SoC, and Subsystem

Specification (Design)

Benchmarks

- High-speed and standard Interfaces like PCIe, CXL, NVMe, Ethernet, UCIe, Display, ARINC
- Automotive & Networking chip interfaces

Key Features

- **Auto-generates** features, test, assertion, and coverage plans directly from specification
- **Auto-generates** UVM-compliant test sequences directly from specification
- **Adapts** dynamically to evolving specifications
- **Auto-embeds** debug data for faster root-cause analysis
- **Integrates** seamlessly with existing verification flows

SdCvP-X™: Verifying Complex Designs

Verification Challenges	SdCvP-X™ Tool
Manual, fragmented planning	Auto-generated, spec-aligned comprehensive segregated verification plans for IP, SOC, & Subsystem directly from specifications
Specification Alignment	Plan evolves with specification + Version traceability
Manual Plan Alignment with Specification and upgrades	Aligns manual plans to eliminate gaps and ensure consistency and boost productivity.
Debug Inefficiencies	Traceability matrix + mapping to features/test points
Configuration Explosion	Smart filtering, collapsing, and prioritization
Reuse across derivatives	Scalable planning across standard & custom designs

Who Benefits: Across the Project Hierarchy

Role	Pain Point	SdCvP-X™ Benefit
Engineers (DV)	Long debug cycles, manual test list creation	Faster test planning + structured debug : Boosts Productivity
Architects	Hard to validate spec implementation	Full traceability from spec to test
Program/Project Managers	Poor visibility on verification progress	Actionable insights, milestone-based views
Business Leaders & Industries	Risk of re-spin, time-to-market delays	Higher First Silicon Success, faster Time to market

SdCvP-X™: GUI & OUTPUT

Design Domain
☒ IP
☐ SoC
☐ Subsystem

Upload Specification

Base Specification (PDF).docx

CXL-3.1-Specification © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED.pdf

Base Register Specification (.xlsx).PDF

Base Add-on Specification (PDF).docx

Base Add-on Register Specification (.xlsx).PDF

Feature List
Do you want specification feature list extraction?
☒ Yes
☐ No

Download Feature List

Do you want to upload modified feature list according to your design?
☐ Yes
☒ No

Verification Plan Filters

Configuration Enabled
☒ Yes
☐ No

Constraints Enabled
☒ Yes
☐ No

Coverage Enabled
☒ Yes
☐ No

Assertion Enabled
☒ Yes
☐ No

Debug Enabled
☒ Yes
☐ No

UVM Test Sequence Enabled
☒ Yes
☐ No

Confirm Filters

Generate Verification Plan

```

transcript.txt
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Domain selected: IP
Uploaded: CXL-3.1-Specification © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED.pdf
Reading Specification: CXL-3.1-Specification © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED.pdf
Created: TEMP_Proc_Data_SdCvP-x
Extracted 16 features.
Feature list saved : /Users/abhilashajha/Desktop/feature_list.xlsx
Verification Plan Filters confirmed:
Configuration Enabled: No
Constraints Enabled: Yes
Coverage Enabled: No
Assertion Enabled: Yes
Debug Enabled: No
UVM Test Sequence Enabled: Yes
Verification plan generated and saved: /Users/abhilashajha/Desktop/verification_plan.xlsx
Deleted: TEMP_Proc_Data_SdCvP-x

```

Verification Plan

UVM Sequence Library

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S.No	Section	Layer/Component	Feature	DUT Type	Testpoint	Scenario	Configuration	Constraint	Assertion	Coverage	Spec Variables

SdCvP-X™: Tiered Product Options

Standard

- ✓ Comprehensive Verification Plan
- ✓ Features, Test Points & Scenarios
- ✓ Configuration & Constraints Plan

Advanced

- ✓ All Standard Features
- ✓ Assertion Plan
- ✓ Enhanced Specification Evolution
- ✓ Testplan Alignment

Most Popular

Premium

- ✓ All Advanced Features
- ✓ Coverage Plan
- ✓ Spec Variables, Debug Info & Traceability
- ✓ Auto-Generated Test Bench
- ✓ Priority Support & Success Manager

Contact Us

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